

steadily. In case there is a signal problem, the signal LED will blink fast.

Enter the following command in the terminal and it will display the output to show the list of active devices where the USB-to-serial dongle/adaptor is mounted:

```
$ ls /dev/ttyUSB*
```

The output will look like /dev/ttyUSB0 or /dev/ttyUSB1 or /dev/ttyUSB2.

Remove and re-insert the dongle and check 'ls /dev/ttyUSB*' to ensure the location.

To send an SMS via Raspi, install an SMS gateway and configure the configuration file for Gammu known as gammurc file (Gammu library configuration).

Type the following command and start preparing gammurc file as follows:

```
$ sudo nano .gammurc
[gammu]
```

Now, configure as shown below:

```
port=/dev/ttyUSB0
connection=at115200
name=gammu
model=
```

Save by pressing ctrl + o and ctrl + x on the keyboard.

Copy this Gammu configuration file to its default location, which is at /etc/

```
$ sudo cp .gammurc /etc/
```

Check whether Gammu is ready for work by typing the following command:

```
$ gammu --identify
```

```
Device: /dev/ttyUSB0
Manufacturer: Undefined
Model: Unknown (M590)
Firmware: Undefined
IMEI: 355911047632749
SIM IMSI: 404929508997756
```

```
pi@raspi40 /var/www/html $
```

This shows that Gammu has found M590 GPRS shield and is now ready for heavy work. In case the shield is not found, check the USB

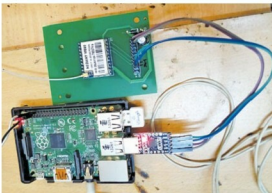


Fig. 3: GPRS shield and dongle connections

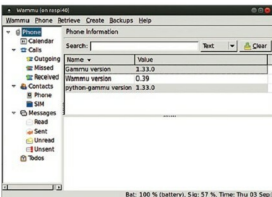


Fig. 4: Wammu graphical interface

port and retry.

Gammu has a graphical interface known as Wammu, which is a mobile phone manager running on Linux, Windows and possibly on other platforms also, where communication is made using Gammu library. Wammu is a very lightweight program. Screenshot of Wammu is shown in Fig. 4.

Type the following command in the terminal and press Enter.

```
$ wammu

Go to Wammu→Settings to select the correct USB port to find the GPRS shield.
```

Go to Phone→Connect and you are now ready for sending/receiving an SMS on Wammu.

Command line interface. Here is how Gammu works from command line. Once you understand the commands given below, running these in PHP is a child's play.

```
$> echo "Hello EFY, how is life?"
j gammu --sendmsg text +91xxxxxxx
```

SMS server design using PHP

In this case, Raspi SMS server is made using PHP script.

There are many third-party open source PHP SMS servers like gammu-php, kalkun, etc, but here we have built a simple PHP SMS server.

PHP script. The small script (sms.php) is written in PHP. It will open in the browser window if Apache and PHP are installed on your Raspi.

Set up Apache Web server on Raspi using the following command:

```
$ sudo apt-get install apache2
php5-dev php5
```

Copy and paste sms.php program into Apache document root directory of Raspi, which is normally /var/www/html or whatever you have set.

For running Gammu super user permission for /usr/bin/gammu.

Open sudoers file in /etc/sudoers and add the following line below it. But sudoers is one of the most sensitive and important file in Wheezy Linux. Any unintended error in this file may corrupt your entire system and that is why it is recommended to edit this file as given below:

```
$ sudo visudo
```

This command will open sudoers file in nano editor.

Add the following line:

```
www-data ALL = (ALL) NOPASSWD: /usr/bin/gammu
```

Close nano editor by pressing control + x. Save it and relax.

www-data or the browser is now capable to run sms.php program.

Write the message in less than 200 words as shown in Fig. 5. Copy and paste the mobile phone numbers separated using commas (,), press Send and your SMSes are